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Flame dynamics

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Abstract

This lecture describes recent theoretical developments associated with the dynamics of flames, obtained primarily by exploiting the various temporal and length scales involved in the combustion process. In premixed flames the focus is on flame–flow interactions that occur during the nonlinear development of hydrodynamically unstable large-scale flames, or during the propagation of curved flames in two-dimensional channels. The second part of the paper deals with non-premixed and partially premixed flames, where the focus is on understanding the nature of diffusive-thermal instabilities including the effect of thermal expansion, and on stabilization mechanisms of edge flames, which possess characteristics of both premixed and diffusion flames. The results presented in this talk illustrate how simplified models, when analyzed to their extreme, yield predictions of qualitative nature with physical insight that have advanced our understanding of combustion. This insight can be used to guide the experimental efforts, explain observations and validate large-scale numerical simulations.
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1. Introduction

Theory has made a strong impact on combustion science and will continue to play an important role in the foreseeable future. The recent advances in experimental methods, in particular laser diagnostic techniques, and in computational capabilities, including the accessibility of computers and the efficiency of numerical tools, are providing combustion scientists with ample data that require understanding and interpretation. Fundamental understanding is best achieved by constructing simplified models that retain the main features of the phenomena being considered and finding solutions to these models, either analytically or numerically, in a manner that makes transparent the physical interactions occurring in the various regions of the flow field and the effects that these have on the overall process.

Modeling combustion systems involves the entire gamut of complexities associated with multi-component flows, heat and mass transfer and chemical reactions, further complicated by a large number of elementary chemical reactions and the multidimensional and time-dependent nature of the flow. Such processes, however, occur over a wide range of temporal and spatial scales. The chemical reactions occurring in mixtures commonly used in combustion devices evolve over disparate time scales that are relatively short compared to the flow or diffusion time scales, which imply that the chemical activity is confined to thin reaction layers. The reaction zones are embedded in the flame, where diffusion, heat conduction and

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viscous dissipation take place. The flame thickness, normally a fraction of a millimeter, is small when compared with the hydrodynamic length scale associated with the geometrical dimensions of the vessel within which combustion occurs. And on a yet larger scale, there is a nonlinear interaction with the acoustic field generated, for example, at the combustor's inlet. Resolving such problems, in a way that faithfully represents the underlying physico-chemical processes on all scales, small and large, is computationally intensive. It is not surprising, therefore, that the techniques of asymptotic analysis are the primary tools that have been used to derive and analyze simplified mathematical models of flames.

Perturbation or asymptotic methods are used to replace a complicated problem by a simpler one, taking advantage of the relative magnitude of the different controlling parameters and/or the disparate scales in the problem. The neglected terms are clearly delineated and, in principle, can be incorporated in higher order corrections by systematically constructing a scheme of successive approximations with improved results. Such an approximation is referred to as a *rational approximation* and is one of the most effective tools available to scientists and engineers for constructing reliable solutions that provide deep physical insight. Van Dyke [1] provides a remarkable list of very useful approximations in fluid mechanics that he calls *irrational approximations*, namely that are not deduced by a limit process expansion. In contrast to asymptotic methods, the success of such an approach is not known a-priori and there is no clear process by which the results can be improved. Attempting such an approximation requires ingenuity and deep physical insight and therefore should not be discarded a-priori. Its merit clearly lies on whether it produces sensible results that compare favorably with experiments. Often the asymptotic nature of an ad-hoc approximation is revealed in future studies. An example in combustion is the infinitely fast chemistry limit used in diffusion flames, which was first introduced by Burke and Schumann [2]. The recognition that the solution corresponds to the asymptotic limit of an infinite Damköhler number, namely when the chemical reaction time is much shorter than the flow time, was discovered much later. This systematic approach also identifies the relative thickness of the reaction zone and leads to an appropriate formulation of its internal structure with solutions that smoothly connect the discontinuous slope across the sheet.

Because of the complexity and the highly nonlinear nature of the governing equations of combustion problems, simplifications that appear quite dubious a-priori are often introduced. In studying diffusion flames in concentric ducts, Burke and Schumann [2] assumed that the axial mass flux throughout the duct remains constant. They effectively introduced a constant-density approximation in order to decouple the hydrodynamic from the transport equations. The constant-density formulation can be systematically obtained as an asymptotic limit when the heat-release parameter is small. In real systems, however, the heat release is never small and density varies significantly across the flame. Nevertheless, the approach is useful in elucidating some of the intricacies associated with thermo-diffusive aspects of the combustion process. The flame shapes and heights predicted by Burke and Schumann [2], for example, are in remarkably good agreement with experimental results. The stability of a planar premixed flame is another example where the constant-density approximation has been used to filter out hydrodynamic disturbances and provide unequivocal results about the role of diffusion on the onset of instabilities in the form of cells and/or oscillations [3]. These results were instrumental later on in constructing a more comprehensive stability theory that includes the previously missing hydrodynamic effects [4–6], previously discovered by Darrieus and Landau.

Asymptotic treatments of combustion problems lead to simplified models that occasionally can be solved analytically, but otherwise require a numerical approach. An argument often expressed is that since a numerical solution is sought, one should then use the more complete governing equations that include detailed chemistry and transport. This suggestion is shortsighted. Even if a reliable numerical strategy is available and integration over a sufficiently long time is possible, which is rare in combustion problems, the inclusion of everything that is likely to be of quantitative importance will significantly increase the number of parameters in the model and will not necessarily provide better understanding. The simplified problem possesses several advantages when the goal is to identify the key physics underlying a particular phenomenon. It is usually not as numerically intense as the complete problem, and since it contains fewer parameters, it is possible to obtain full parametric solutions rather than solutions that are restricted to only a set of parameters. It also provides deeper physical insight by more easily identifying cause and effect. In addition, asymptotic solutions may be useful in guiding and validating large-scale computations of the more complete governing equations. In this respect, asymptotic and numerical methods complement each other.

A prominent theme in this lecture will be to describe how reduced models, when analyzed to their extreme without being distracted by the adopted simplifications, provide physical insight that have advanced our understanding of combustion. This insight has been used to guide experimental efforts, explain observations and validate large-scale numerical simulations. The problems

discussed below are concerned with the dynamics of premixed flames, non-premixed (diffusion) and partially premixed flames. The paper, however, is not a review on flame dynamics; it complements related talks given at previous symposia [7–9], with a selection of topics that reflects the author's own interest. References are only cited when directly relevant to the discussion, with no attempt to exhaustively cover all the pertinent literature on the subject.

2. Premixed flames

When a combustible gas mixture is ignited by a heat source within a flow field, a deflagration wave, or premixed flame, propagates relative to the flow into the unburned gas consuming the reactants and leaving behind hot products. The simplest problem in premixed combustion is the planar unconfined flame that propagates steadily into a quiescent mixture. The flame then depends on two parameters, which are characteristics of the combustible mixture: the adiabatic flame temperature T_a, achieved when the mixture burns to completion under isobaric and adiabatic conditions, and the laminar flame speed S_L, which is the speed that the (planar) flame travels into the mixture in the absence of heat losses. In combustion applications flames seldom behave in such an idealized way; they are typically corrugated and propagate in a non-steady manner. The temperature and propagation speed vary along the flame front and depend on nonuniformities in the mixture composition, heat losses, the diffusivity of the various species comprising the mixture and the local flow conditions ahead of the flame.

It is commonplace in theoretical studies of combustion problems to describe the complex reaction scheme involved in a particular fuel oxidation by one-step Arrhenius kinetics. The adiabatic flame temperature obtained from an overall energy balance is then given by $T_a = T_u + QY_u/c_p$, where T_u is the temperature of the fresh unburned gas, Y_u the mass fraction of the deficient reactant in the fresh mixture (fuel in a lean and oxidizer in a rich mixture), Q the heat released per unit mass of (deficient) reactant and c_p the specific heat of the mixture (at constant pressure). The determination of the flame speed requires solving the appropriate mass and energy conservation equations. But despite the one-dimensional simplification, exact solutions of these highly nonlinear equations are not available. Approximate expressions for the flame speed date back to Mallard and Le Chatelier [10], but a significant breakthrough was due to Zel'dovich and Frank-Kamenetsky [11] who, using physical arguments, obtained an expression for the flame speed valid for large activation energy. The realization more than 30 years later that the problem can be solved formally using matched asymptotic expansions [12], and the subsequent review article by Williams [13] that advocated the use of this formal approach, paved the way to much of the theoretical development that has taken place in recent years.

The asymptotic treatment considers that both, the Damköhler number representing the ratio of the flow to the chemical reaction times and the non-dimensional activation energy, or Zel'dovich number, are large parameters related by an appropriate distinguished limit. As a consequence, chemical reaction is confined to a thin layer that separates the preheat and equilibrium zones, and is negligible elsewhere. The reaction diffusion balance yields, through asymptotic matching, jump conditions across the reaction zone that enable an explicit solution across the entire flame including an expression for the laminar flame speed. The formal approach permits deriving more accurate expressions for S_L by retaining higher order terms in the expansions. More importantly these results can be generalized and used, through multi-parameter expansions and appropriate distinguished limits, to describe the structure, speed and temperature of multi-dimensional and time-dependent flames [14,15,5]. This has been an essential development in advancing the hydrodynamic theory of flame propagation, which in its primitive form dates back to Darrieus [16] and Landau [17], with implications on numerous areas of combustion research.

2.1. Hydrodynamic theory

The mathematical description of multi-dimensional premixed flames can be presented at various levels of detail. Viewed on the representative fluid-flow length scale, the whole flame, where chemical reactions, diffusion, heat conduction and viscous dissipation take place, is relatively thin. The flame may then be considered an interface that separates the unburned fresh mixture from the hot combustion products. The flow field on either side of the flame sheet, whether laminar or turbulent, is determined by hydrodynamic equations and must satisfy appropriate jump conditions across the sheet, consistent with the conservation laws of mass, momentum and energy. The hydrodynamic model was first introduced by Darrieus [16] and Landau [17] who examined in this manner the stability of a planar flame front. They solved the Euler equations on either side of the perturbed flame surface, and connected the solutions on each side by using the Rankine–Hugoniot relations. To complete their formulation, they assumed that the flame sheet propagates at a constant speed relative to the flow immediately ahead of the flame surface. The Darrieus–Landau model ignores the transport phenomena and thus fails to properly account for the influence

of diffusion on the propagation. A great deal of work was done in the years that followed to improve on this model; notable is the study of Markstein [18] who assumed a dependence of the flame speed on the local curvature of the flame front through a phenomenological constant that has become known as the Markstein length (or Markstein number, when expressed in units of the diffusion length).

The more rigorous treatments [5,6,4] exploit the multi-scale nature of the problem characterized by two disparate length scales: the diffusion length scale l_f representing the flame thickness, where $l_f = \mathscr{D}_{th}/S_L$ with $\mathscr{D}_{th}$ the thermal diffusivity of the mixture, and the hydrodynamic length scale L associated, for example, with the average size of the wrinkles on the flame front or with the geometrical dimensions of the vessel within which combustion takes place. In the limit $l_f/L \to 0$, the flame is confined to a sheet on either side of which the flow is described by solving the incompressible Navier–Stokes equations with different densities and viscosities, ρ_u, μ_u for the unburned and ρ_b, μ_b for the burned gas, respectively. The internal flame structure is resolved on the diffusion scale with asymptotic matching providing expressions that relate the pressure and velocities across the entire flame, and an equation for the speed, or equivalently for the instantaneous shape of the flame front. The jump conditions take the form of generalized Rankine–Hugoniot relations, namely with corrections on the order of the flame thickness that account for accumulation of mass and momentum inside of, and transverse mass and momentum fluxes through, the thin flame zone. The flame speed depends on the local curvature of the surface and on the underlying hydrodynamic strain, effects that are said to *stretch* the flame surface. The entire formulation is then a nonlinear free-boundary hydrodynamic problem with conditions at the free surface that describe the influences of the diffusion processes occurring within the flame zone. The theory accounts for density variations, effects of differential and preferential diffusion with temperature-dependent transport coefficients, and effects due to stoichiometry and arbitrary reaction. In its most general form the mathematical formulation was given by Matalon et al. [19]. Here, we only comment on selected flame characteristics; further details can be found in the referenced paper.

The flame speed is unambiguously defined in the present context as the propagation velocity of the sheet relative to the unburned gas. It is given by

$$S_f = S_L - \mathscr{L}\mathbb{K} \tag{1}$$

where $\mathscr{L}$ is the Markstein length and $\mathbb{K}$ is the (local) flame stretch rate. Flame stretch is a measure of the flame front deformation resulting from its motion and nonuniformities in the underlying flow field. It may be defined, in general, as the proportionate rate of change of the surface area of a Langrangian flame element [20]. Within the present approximation it can be expressed as the combined effects of curvature and strain, namely $\mathbb{K} \sim S_L\kappa + \mathbb{K}_s$ where $\kappa = 1/\mathscr{R}_1 + 1/\mathscr{R}_2$ is the total curvature, with $\mathscr{R}_1, \mathscr{R}_2$ the principle radii of curvature, and $\mathbb{K}_s = -\mathbf{n}\cdot\mathbf{E}\cdot\mathbf{n}$ is the hydrodynamic strain, with $\mathbf{E}$ the rate of strain tensor [19]. An equivalent useful formula, expressed in terms of the local gas velocity $\mathbf{v}$ was given by Matalon [21] in the form $\mathbb{K} = V_f\nabla\cdot\mathbf{n} - \mathbf{n}\cdot\nabla\times(\mathbf{v}\times\mathbf{n})$, where V_f the velocity of the front (in a fixed frame of reference) and $\mathbf{n}$ a unit vector normal to the sheet and pointing towards the burned gas region. The dimensionless parameter $l_f\mathbb{K}/S_L$ is often referred to as the Karlovitz number.

The Markstein length, which measures the sensitivity of the flame speed to stretch, is given by

$$\mathscr{L} = \left\{\frac{\sigma}{\sigma-1}\int_1^\sigma \frac{\lambda(\tau)}{\tau}\,d\tau + \frac{\beta(Le_{eff}-1)}{2(\sigma-1)} \times \int_1^\sigma \frac{\lambda(\tau)}{\tau}\ln\left(\frac{\sigma-1}{\tau-1}\right)d\tau\right\} l_f \tag{2}$$

where $\sigma \equiv \rho_u/\rho_b$ is the unburnt-to-burnt density ratio, or thermal expansion parameter, $\beta \equiv E(T_a - T_u)/R^o T_a^2$ is the Zel'dovich number, with E the overall activation energy of the chemical reaction and R^o the universal gas constant, and λ is the (dimensionless) thermal conductivity of the mixture assumed to depend solely on temperature. Here the state of the fresh unburned gas is chosen as reference so that $\lambda = \lambda(\tau)$ with $\tau = T/T_u$. The effective Lewis number of the mixture, Le_{eff}, is a weighted average of the individual Lewis numbers, Le_F and Le_O, representing the ratios of the thermal diffusivity of the mixture to the mass diffusivities of the fuel and oxidizer, respectively. It is given by

$$Le_{eff} = \begin{cases} \frac{Le_O + (1-\tilde{\phi})Le_F}{2-\tilde{\phi}} & \text{lean} \\ \frac{Le_F + (1+\tilde{\phi})Le_O}{2+\tilde{\phi}} & \text{rich} \end{cases}$$

where $\tilde{\phi} \sim \beta(\phi - 1)$ with ϕ the equivalence ratio. For a stoichiometric mixture the effective Lewis number is the average of the individual Lewis numbers of the two reactants. For an off-stoichiometric mixture the deficient component is more heavily weighted such that for very lean/rich mixtures the effective Lewis number is practically that of the fuel/oxidizer, respectively. Equation (1) implies that a positively-stretched element of the flame surface propagates at a slower speed than S_L if $\mathscr{L} > 0$, namely when Le_{eff} is sufficiently large as for lean hydrocarbon–air mixtures (except possibly for methane), and at a faster speed than S_L if $\mathscr{L} < 0$ or when Le_{eff} is sufficiently small as for rich hydrocarbon–air mixtures. The reverse is true for a negative-stretched element of the flame surface.

The Markstein length varies significantly with the mixture's composition, as shown in Fig. 1. The graph displays the dependence of the Markstein number $\mathscr{L}/l_f$ on equivalence ratio for different fuel–air mixtures, calculated from Eq. (2) with $\lambda = \tau^{1/2}$. For alcohol– and hydrocarbon–air mixtures the Markstein number is generally positive and decreases monotonically as the mixture varies from lean to rich conditions. The large variations among the various fuels for lean conditions is due to the difference in their diffusive properties. For rich conditions the effective Lewis number is practically that of oxygen and the relatively minor difference in the Markstein number among the various mixtures is due to the different values of heat release. For mixtures of relatively light fuel, such as methane–air and hydrogen–air mixtures, $\mathscr{L}$ is found to decrease as the mixture varies from rich to lean conditions, and may become negative at sufficiently low values of the equivalence ratio.

It should be noted that the Markstein length is *uniquely* defined in the asymptotic limit considered here, *i.e.* when the whole flame shrinks to a surface that coincides with the reaction sheet. Real flames have a finite thickness and the Markstein length is only known to within an amount proportional to the flame thickness. This requires careful consideration when attempting to compare the theoretical results with experimental data. Since measurements are typically taken at a specific reference location inside the flame, the Markstein length $\mathscr{L}$ must be re-adjusted for this displacement as discussed in [22,23]. The magnitude of $\mathscr{L}$ decreases as a result of this adjustment and for a reference location further upstream, methane–air mixtures may also have negative values of $\mathscr{L}$ in sufficiently lean mixtures. A comparison between these predictions and representative experimental data was presented in [23].

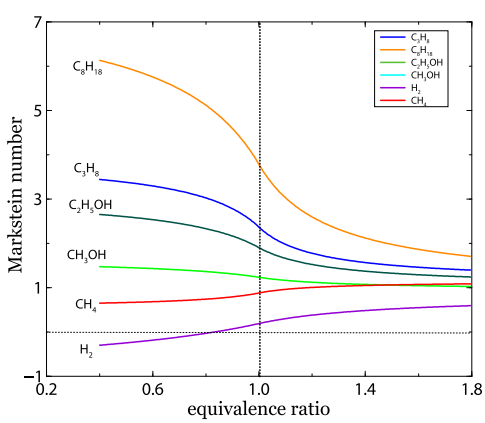

Fig. 1. The dependence of the Markstein number $\mathscr{L}/l_f$ on equivalence ratio for selected fuel–air mixtures, with the reference location taken at the reaction sheet.

Variations of the flame temperature T_f from the adiabatic value are also proportional to stretch, but the deviation depends now on the Lewis number of the *limited* component in the mixture. For a lean mixture

$$T_f = T_a - T_u(Le_F - 1)\left\{\int_1^{\sigma} \frac{\lambda(\tau)}{\tau} \ln\left(\frac{\sigma-1}{\tau-1}\right) d\tau\right\} \frac{l_f}{S_L}\mathbb{K}; \tag{3}$$

in a rich mixture the Lewis number is replaced by Le_O. These variations, however, are relatively small, being not only proportional to l_f, but also to the deviation of the Lewis number from one. The temperature of methane–air flames, for which the Lewis number is always near one, is close to the adiabatic temperature for all stretch rates. For slowly or highly diffusive fuels the temperature of positively stretched elements of the flame surface is slightly lower or slightly higher than T_a, respectively. Under near-stoichiometric conditions it is possible for a fast-diffusing reactant initially in excess to be depleted first in the reaction zone, and hence to become the rate-controlling species; these refinements are discussed in [24].

Measurements of speed and temperature of stretched flames were found to correlate well with expressions (1)–(3) for uniformly-stretched twin flames in a counterflow [25,26], spherically expanding flames [27,28] and Bunsen burner flames [29,30], and were used to experimentally determine the Markstein length that quantifies stretch effects [31–33].

The hydrodynamic model exhibits a complex interaction between the flame and the surrounding gas and retains much of the physics of real flames. The model is an appropriate framework in which to study the development of instabilities and the evolution of large-scale corrugated flames. This asymptotic theory has also spurred activities in a number of other areas. It provided the basis of the flamelet concept that views a turbulent flame as an ensemble of stretched laminar flamelets embedded in a turbulent flow, and helped identify the different regimes of premixed turbulent combustion, including the corrugated-flamelets regime where the internal flame structure is practically unaffected by the turbulence, and the thin-reaction-zone regime where small eddies of the size of the Kolmogorov scale are able to penetrate the preheat zone but not the reaction zone. It has also inspired modeling efforts of turbulent flames using the so-called G-equation, which involves expressing Eq. (1) in terms of a function $G(\mathbf{x}, t) = 0$ that describes the instantaneous shape and location of the flame sheet [34,35].

There are limitations of the hydrodynamic theory that are encountered when trying to model flames in real flows. The theory is restricted to weakly-stretched flames and does not predict quenching at high strain rates. It requires the

flame surface to consist of a single sheet, with no holes or folds. Furthermore, diffusion influences, which have been added as a perturbation, are not properly resolved in regions where, to leading order, there is a discontinuity in the slope of the flame front; for example near the tip of a Bunsen flame or where the flame comes in contact with a wall. The formulation in such cases is incomplete and must be supplemented with additional conditions that properly mimic the local description. Attempts have been made to incorporate such effects in an ad-hoc manner, but more work is needed in this area. Recently, the theory has been generalized to account for volumetric heat losses with provisions for local quenching when the total integrated losses from the flame zone are excessive [36]. The resulting formulation for the flow field remains unchanged if the flame speed is normalized by the speed of a planar non-adiabatic flame and lengths are scaled accordingly. The only difference is that the Markstein length now admits two separate values, one associated with curvature and the other with strain, values that can vary significantly near the extinction conditions. Another drawback that may be found when trying to relate the theoretical results to real flames is the one-step Arrhenius kinetics adopted in resolving the internal flame structure. In rich hydrocarbon flames, for example, the fuel may be partially oxidized to CO and H_2 with no oxygen left, leading to a Markstein length that may differ from that predicted in Fig. 1. Although flame structure analyses with reduced chemistry models have been successful in reproducing aspects of real flames [37–40], they have been limited to simple flow configurations with no attempt made to examine the influence of chemistry on flame dynamics. Possible extensions of the hydrodynamic theory in this and other directions have been already noted in a recent review [9] prepared in honor of the 50th anniversary of the Combustion Institute.

2.2. Linear stability analysis

Hydrodynamic models are extremely useful in studies aimed at unraveling the intricate nature of flame instabilities. The complex nature of the mathematical stability problem, which requires examining the response of the flame to *arbitrary* initial conditions, disturbances of *all possible wavelengths*, and a whole range of physical parameters is a formidable task that cannot be easily accomplished by numerical means. The relative simplicity of the hydrodynamic model yields analytical expressions for the dispersion relation that describes the growth of arbitrarily small disturbances, and since the model contains parameters that mimic the various physico-chemical processes involved, the influence of these parameters on stability is clearly delineated.

2.2.1. Planar flames

For a nominally planar flame the dispersion relation that describes the growth ω of a disturbance with transverse wavenumber k is given by

$$\omega = \omega_{\mathrm{DL}} S_{\mathrm{L}} k - l_{\mathrm{f}}[B_1 + \beta(Le_{\mathrm{eff}} - 1)B_2 + PrB_3]S_{\mathrm{L}} k^2, \tag{4}$$

$$\omega_{\mathrm{DL}} \equiv \frac{-\sigma + \sqrt{\sigma^3 + \sigma^2 - \sigma}}{\sigma + 1}.$$

Here Pr is the Prandtl number and the coefficients B_1, B_2, B_3, which depend solely on σ, are all positive. Explicit expressions for these coefficients can be found in [19].

The first term in (4) was obtained by Darrieus [16] and Landau [17] in an analysis that ignores the flame structure (practically assuming $l_{\mathrm{f}} \to 0$), and does not account for diffusion processes at the small scale. The conclusion that planar flames are unconditionally unstable, is known as the hydrodynamic or the Darrieus–Landau (DL) instability. The physical origin of the instability, which is due to thermal expansion, is illustrated in Fig. 2. If the planar front is slightly perturbed, such that $y = -S_{\mathrm{L}} t + \varphi(x, t)$, the Rankine–Hugoniot relations across the flame together with the constant flame speed hypothesis imply that

$$v^+ = v^- = \varphi_t \qquad u^+ - u^- = -(\sigma - 1)S_{\mathrm{L}}\varphi_x \tag{5}$$

where v, u are the induced velocities in the axial and transverse directions y, x, respectively, and the superscript $\mp$ relate to conditions on the unburned/burned side of the unperturbed sheet. The first of these relations states that the flame front is convected by the flow; the latter shows that for $\sigma > 1$ the jump in the transverse velocity component is as shown in Fig. 2(a). For weak thermal expansion the generation of vorticity in the burned gas region is negligible and the flow ahead as well as behind the flame is irrotational. Vorticity is only concentrated along the sheet so that the perturbed flame is equivalent to a flat vortex sheet of nonuniform strength. The distribution of vorticity is oriented such as to increase the initial flame displacement as shown in Fig. 2(b). The vorticity generated in the burned gas, which for realistic values of thermal expansion may not be neglected, does not have an adverse effect on the motion of the sheet, as the stability analysis shows *a posteriori*.

The second term in (4), which is proportional to l_{f} was obtained more recently [5,6,4]; the three elements in this expression describe respectively the influences of thermal, molecular and viscous diffusion occurring within the flame on stability. Thermal diffusion, which tends to smooth out temperature differences, always has a stabilizing influence. Due to the large change in viscosity across the flame, viscous diffusion also has a stabilizing influence. On the other hand, the effect of molecular diffusion depends on the mixture com-

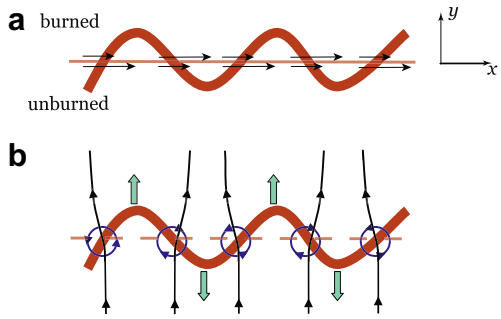

Fig. 2. Schematic showing the physical origin of the DL instability. The jump in the transverse velocity across the flame shown in (a) implies that the perturbed flame is equivalent to a flat vortex sheet of nonuniform strength with vorticity orientation, as shown in (b), that causes an increase in the initial flame displacement. Also shown in (b) is the deflection of streamlines that results from the gas expansion and causes an expansion/contraction of the streamtubes and consequently a pressure gradient that increases an initial displacement of the flame.

position, or the effective Lewis number of the mixture.

For sufficiently small values of the Lewis number, or specifically for $Le_{\mathrm{eff}} < Le^{*}_{\mathrm{eff}}$, the second term in (4) becomes positive, implying that short wavelength disturbances are amplified and the hydrodynamic instability is enhanced by diffusion effects. Thus, even when the long wavelength disturbances are excluded because of the finite size of the burner, or damped as a result of gravity for downward propagation,[1] diffusion effects do not permit the flame to remain flat. This limit is referred to as the diffusive-thermal instability. Proper description of the resulting cellular flame requires including higher order terms in the dispersion relation (4) for short wave stabilization. This was explored for weak thermal expansion, $\sigma - 1 \ll 1$, giving rise to a dispersion relation [41] of the form

$$\omega = \frac{1}{2}(\sigma - 1)S_{\mathrm{L}}k - \frac{\beta}{2}S_{\mathrm{L}}l_{\mathrm{f}}\left(Le_{\mathrm{eff}} - Le^{\mathrm{cr}}_{\mathrm{eff}}\right)k^2 - 4\mathscr{D}_{\mathrm{th}}l_{\mathrm{f}}^2k^4 \tag{6}$$

with the critical Lewis number, $Le^{\mathrm{cr}}_{\mathrm{eff}} \sim 1 - 2\beta^{-1}$, slightly less than one. A similar expression for arbitrary σ is not available, but when $Le^{\mathrm{cr}}_{\mathrm{eff}}$ is estimated from Eq. (4) for the realistic value $\sigma \approx 6$ it yields much lower values, rarely exceeding 0.5. This makes most common combustion mixtures inaccessible to the instability except perhaps for lean hydrogen–air flames [42]. The critical Lewis number $Le^{\mathrm{cr}}_{\mathrm{eff}}$ increases when heat losses are incorporated in the analysis and is closer to one near the flammability limit [43,44]. Cellular flames resulting from diffusive-thermal effects may therefore be observed at near-extinction conditions, for example near burners where heat losses are unavoidable.

When the effective Lewis number $Le_{\mathrm{eff}} > Le^{*}_{\mathrm{eff}}$, which is clearly attainable in lean hydrocarbon–air or rich hydrogen–air mixtures, diffusion acts to stabilize the short wavelength disturbances. Stable planar flames may then result provided the long wavelength disturbances are damped by gravitational forces [17], stretched by hydrodynamic strain [45], or excluded because of the presence of walls in the transverse direction. When the long wavelength disturbances are not totally suppressed a corrugated flame results. The instability, which is predominantly observed in large-scale flames where the stabilizing effects of diffusion are limited, is due to the hydrodynamic instability.

2.2.2. Spherical flames

The onset of cells on spherically expanding flames more clearly illustrates the distinction between the diffusive-thermal and hydrodynamic instabilities. The problem was first examined by Istratov and Librovich [46] who considered a DL model with a Markstein correction, and later by Bechtold and Matalon [47] who incorporated hydrodynamic and diffusion effects in a more systematic way. Let the perturbed front of a spherically expanding flame be expressed in the form $r = R(t)\{1 + A(t)\mathbb{S}_n(\theta, \varphi)\}$, where R denotes the radius of the unperturbed spherical flame, A the amplitude of the disturbance and $\mathbb{S}_n$ the spherical surface harmonics with n an integer, the growth rate is given by

$$\frac{1}{A}\frac{\mathrm{d}A}{\mathrm{d}t} = \frac{\dot{R}}{R}\left\{\tilde{\omega}_{\mathrm{DL}} - \frac{l_{\mathrm{f}}}{R}\left[\tilde{B}_1 + \beta(Le_{\mathrm{eff}} - 1)\tilde{B}_2 + Pr\tilde{B}_3\right]\right\} \tag{7}$$

with coefficients $\tilde{\omega}_{\mathrm{DL}}$ and $\tilde{B}_1, \tilde{B}_2, \tilde{B}_3$ that depend on σ and n, and with the exception of the low values of n are all positive [47]. Consistent with the hydrodynamic description this result is valid for $R > R_0$, where R_0 is the initial flame radius assumed larger than the diffusion length l_{f}. The first term $\tilde{\omega}_{\mathrm{DL}}$ represents the destabilizing effect of thermal expansion, whereas those multiplying l_{f} are, respectively, the influences of thermal, molecular and viscous diffusion.

When $Le_{\mathrm{eff}} < Le^{*}_{\mathrm{eff}}$, a value that slightly differs from the corresponding value for a planar flame, the amplitude grows in time for all n, starting at $t = 0$. The instability must have developed when the flame radius was smaller than the initial radius R_0, and is diffusive-thermal in nature. Indeed, spherically expanding flames in rich hydrocarbon–air or lean hydrogen–air mixtures, character-

[1] For downward propagation, the modification of the DL results shows that gravity acts to stabilize the long wavelength disturbances, disturbances with wavelength $\lambda > 2\pi\sigma S_{\mathrm{L}}^2/g$.

ized by a Lewis number sufficiently less than one, were observed to take on a cellular appearance shortly after ignition [48,49]. When $Le_{\mathrm{eff}} > Le^*_{\mathrm{eff}}$, the right-hand side of (7) for a given n changes sign when the flame reaches a critical size $R = R_c$. All disturbances are initially damped out: the hydrodynamic instability is suppressed because of the large curvature of the front, and since for sufficiently large Le_{eff} molecular diffusion exerts stabilizing influences on the short wavelength disturbances, a smooth (stable) flame results. As the flame grows bigger, namely for $R > R_c$, the flame is hydrodynamically unstable. Indeed, in lean hydrocarbon–air or rich hydrogen–air mixtures, the expanding flame was seen to remain smooth during the early stages of propagation, but took on a cellular appearance once it reached a critical size [50,51]. This has been more clearly illustrated in experiments carried out at different pressure levels [52], as shown in Fig. 3. At five atmospheres a spherically-expanding rich hydrogen–air flame, which is diffusively stable, remains smooth for a significant time after ignition. Under similar conditions but at twenty atmospheres the onset occurs at a much earlier time. At high pressure the flame is thinner, the stabilizing influences of diffusion are minimized and the hydrodynamic instability is enhanced.

Information about the size of the cells that are likely to be observed during the initial stage of the self-wrinkling phenomenon can be deduced from the linear theory. Figure 4 shows the range of unstable modes as a function of the Peclet number $Pe = RS_L/\mathscr{D}_{\mathrm{th}}$ or instantaneous flame size. The nose of the peninsular instability determines the critical conditions for the onset of the instability. Since the fastest growing disturbance corresponds to $n_c \approx 14$, the flame surface at the onset is instantaneously covered with a large number of cells. As the flame continues to expand, modes of higher wavenumber n grow faster and more and more cells develop on the flame surface. Expected cell sizes fall in the range $\lambda_{\min} < \lambda < \lambda_{\max}$ with the larger corrugations stabilized by stretch and disturbances with shorter wavelength stabilized by diffusion. The smallest cell size, $\lambda_{\min} = 2\pi R/n \approx 2\pi l_f$, is nearly a constant that depends on mixture composition. The largest expected cell size, $\lambda_{\max} \approx 2\pi R/n^*$ with $n^* \approx 7$ is proportional to R. This implies that as the flame expands the stretch rate decreases and cells of larger and larger size are able to develop on the flame surface. Large cells, however, are hydrodynamically unstable and have the tendency to split into smaller cells. They are also sensitive to external noise, which, when amplified as a result of the hydrodynamic instability, leads to the spontaneous appearance of small-scale structures, or smaller-sized cells, superimposed on the flame surface.

The stability results have provided a framework for interpreting experimental observations [53,33,54]. The earlier results implied that the theory underestimates the critical Peclet number corresponding to the appearance of cells, which has prompted several improvements of the model [55,56] with results that are more commensurate with the experimental record. It should be noted that the instability threshold is sensitive to the physical parameters, so that a quantitative agreement between theory and experiment requires a sensible choice of flame properties that depend on the local experimental conditions. The evaluation of the thermal diffusivity of the mixture, for example, is subject to uncertainties in the choice

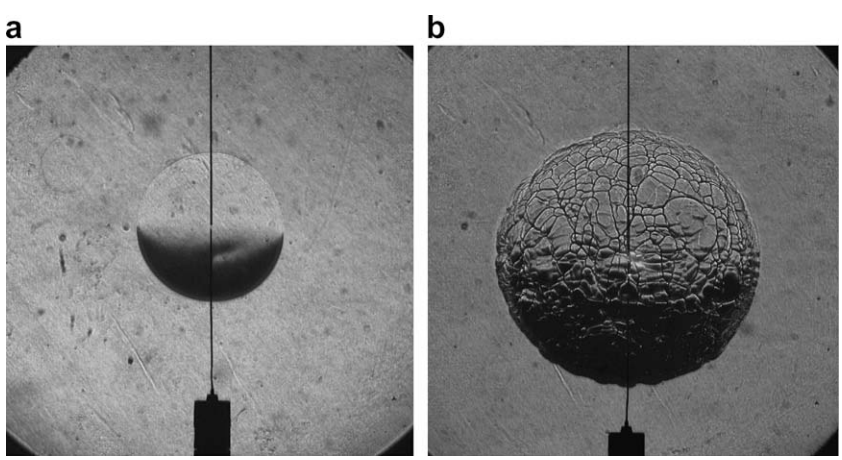

Fig. 3. Photographs of spherically expanding flames in rich hydrogen–air mixtures at (a) 5 atm and (b) 20 atm, respectively, captured 2 ms after ignition. In rich H_2–air mixtures the effective Lewis number is larger than unity and the flames are diffusively stable. The wrinkling of the flame surface is a manifestation of the hydrodynamic instability which, at high pressure, is observed at an earlier time because of the minimized influences of diffusion. The apparent increase in propagation speed is the cumulative result of high pressure and developing instabilities. Courtesy of C.K. Law.

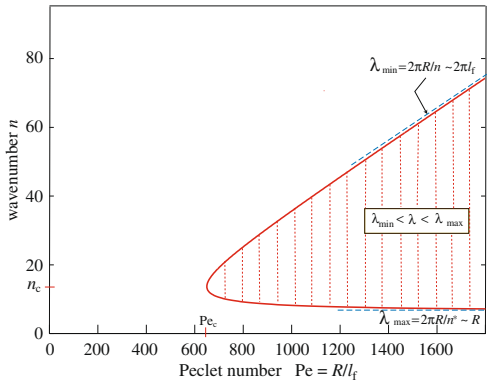

Fig. 4. Peninsula of instability for spherically expanding flames. The shaded region correspond to the range of unstable modes for a given Peclet number, Pe, or flame size; corresponding to $Le_{\text{eff}} = 1.23, Pr = 0.7, \sigma = 6$ with $\lambda = \tau^{1/2}$. The critical value Pe_c, which marks the onset of the self-wrinkling phenomenon, assumes a smaller value when the mixture is made richer.

of temperature and concentrations, as is the global activation energy, which additionally depends on the detailed reaction mechanism. With a judicious choice of the parameters the predicted critical Peclet number was found to agree with the experimental conditions under a wide range of conditions [57,58]. Regarding cell sizes, the experimental data determined from flame visualization and Planar Laser-Induced Fluorescence images of cells [54] fall generally within a peninsular of instability, similar to that of Fig. 4.

2.3. Nonlinear evolution of hydrodynamically unstable flames

The linear stability results identify the critical conditions for the inception of instabilities, and clarify the physical mechanisms responsible for their onset. Beyond the instability threshold, when the unstable modes have grown to sufficiently large amplitudes, nonlinear effects may no longer be neglected. The hydrodynamic instability is responsible for the formation of sharp crests that are often seen pointing toward the burned gas [59,60] or the roughened surface observed on centrally-ignited expanding flames [61, p. 432], the description of which must be based on a nonlinear analysis.

Theoretical progress on the nonlinear development of hydrodynamically unstable flames ($Le_{\text{eff}} > Le^*_{\text{eff}}$) has relied primarily on simplified models of a nominally planar front. One such model is the Michelson–Sivashinsky (MS) equation [41,62] obtained in the weakly-nonlinear long wave asymptotic limit, when the density difference across the flame is small. Attempts to extend the model by including higher order terms [63,64], produced results with only minor differences. With the flame front described by $y = -S_L t + \varphi(\mathbf{x}, t)$, the flame displacement satisfies the integro-differential equation

$$\frac{\partial \varphi}{\partial t} + \frac{S_L}{2}(\nabla\varphi)^2 - S_L \mathscr{L} \nabla^2 \varphi - \frac{\sigma - 1}{8\pi^2} S_L \iint |\mathbf{k}| e^{i\mathbf{k}\cdot(\mathbf{x}-\tilde{\mathbf{x}})} \varphi(\tilde{\mathbf{x}}, t)\, d\mathbf{k}\, d\tilde{\mathbf{x}} = 0.$$

On a finite domain of characteristic size L the problem depends on a single parameter $\alpha = \mathscr{L}/(\sigma - 1)L$, or the reduced Markstein number.

The MS equation with periodic boundary conditions admits exact solutions, known as pole-solutions [65], which correspond to cusp-like structures (or cells) extended periodically in the transverse directions that propagate at a constant speed without change in shape. In two-dimensions they take the form

$$\varphi = -U(\sigma - 1)^2 t + \Phi(x), \quad U = \frac{S_L}{2L} \int_0^L \Phi_x^2\, dx.$$

so that the incremental increase in propagation speed is proportional to the fractional increase in surface area of the flame front. For $\alpha > 0$ there always exists a pole solution which is asymptotically stable [66,67]. The zero-pole solution corresponds to a planar front and is the only stable solution for $\alpha < 2$. Higher order pole solutions have similar shapes but higher amplitudes and sharper peaks. Numerical integration of the MS equation shows that, indeed, short-wavelength corrugations introduced through initial disturbances merge forming bigger and bigger cells that eventually coalesce into a single-peak structure filling the entire interval [62,68]. This solution, which coincides with the corresponding pole solution for the given value of α, continues to propagate at a constant speed as time progresses with no change in shape. The solution is easily extended to three-dimensions, where the flame surface $\Phi(x, z)$ is now the superposition of two pole solutions [69]. The propagation speed U in this case is higher than the corresponding two-dimensional case.

Numerical simulations of the Navier–Stokes equations, without restriction on the thermal expansion parameter, were carried out recently within the framework of a hydrodynamic theory [70,71]. Figure 5 shows the long-time development of a perturbed planar front at consecutive time intervals, for $\alpha = 0.0025$ and the realistic value $\sigma = 6$. Similar to the predictions of the MS equation the flame evolves into a single-peak structure that spans the entire interval and propagates at a constant speed. However, with realistic values of σ the flame reaches a larger amplitude and propagates at a higher speed. The incremental increase in speed is shown in Fig. 6 as a function of $1/\alpha$

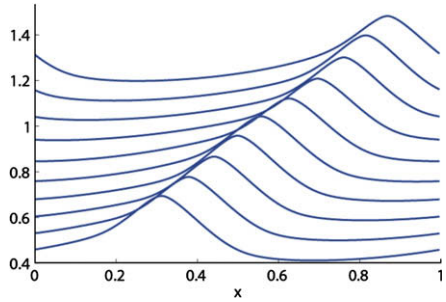

Fig. 5. The nonlinear evolution of a perturbed planar front for $\alpha = 0.0025$ with a realistic $\sigma = 6$. The flame profiles $\varphi(x)$, plotted at consecutive times from top to bottom, correspond to the long-time behavior of the solution after the initial transient has faded out. The steadily-propagating single-peak structure that develops is a consequence of the hydrodynamic instability.

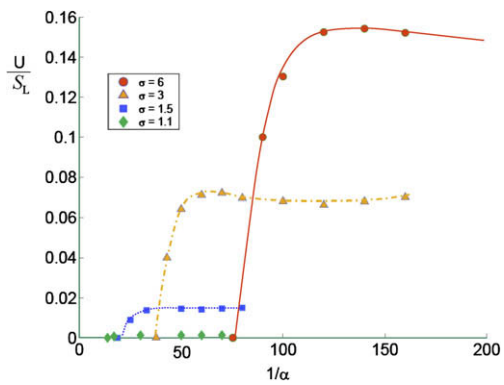

Fig. 6. The incremental increase in speed of the single-peak structure resulting from the hydrodynamic instability as a function of the parameter $1/\alpha$ for increasing values of the density ratio σ.

for increasing values of the density ratio σ. The intersection of these curves with the axis marks the critical value of $1/\alpha = (\sigma - 1)L/\mathscr{L}$, above which the planar flame is hydrodynamically unstable. In narrow domains the planar front is the preferred mode of propagation, but in larger domains a single-peak structure similar to the one shown in Fig. 5 is the favored shape. The incremental increase in speed of the evolving flame rises sharply beyond the bifurcation point and tends rapidly to an asymptotic value, which, for $\sigma = 6$, corresponds to a 15% increase. This behavior is qualitatively captured by the MS equation when scaled appropriately. The incremental increase in speed varies linearly with thermal expansion [71]. The hydrodynamical instability, therefore, leads to corrugated structures with transverse dimensions that are much larger than those observed in the more common laboratory cellular flames, which are a consequence of a diffusive-thermal instability. Moreover, the structure and propagation speed of the evolving flames are practically independent of the domain size and mixture composition.

When computations are carried out in sufficiently large domains, so that α is relatively small, the general structure of the solution is retained but the evolving flames do not settle to a steadily propagating state. Rather, small-scale wrinkles appear sporadically on the flame surface, as shown in Fig. 7, and the speed of propagation varies continuously in time. The wrinkles appear first on the troughs (where the curvature is small), propagate along the flame surface and disappear at the crests. This unsteady behavior, found in both the MS as well as in the fully nonlinear Navier–Stokes models is not a self-sustained phenomenon, but rather a response to an ever-present background noise (*e.g.* numerical noise). Indeed, the appearance of small-wrinkles can be reduced significantly, or eliminated, if the numerical accuracy is increased. Moreover, small-scale wrinkles that closely resemble those emerging from numerical noise can be artificially created on a smooth profile of the MS equation, by introducing a low amplitude external forcing that represents random noise. In real flames external noise may result, for example, from a weakly turbulent flow in the incoming stream. The turbulence then provides a permanent level of noise which, when amplified by the hydrodynamic instability, results in small-scale wrinkles that "contaminate" the flame surface. The appearance of a wrinkle leads to an increase in surface area and consequently an increase in propagation speed until it merges at the crest. The disappearance of a wrinkle is associated with a decrease in propagation speed which continues until new wrinkles reappear on the

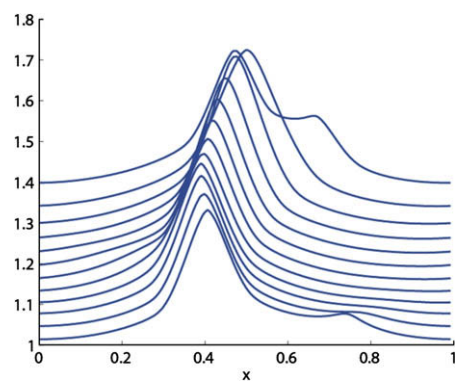

Fig. 7. The long-time development of a perturbed planar front for $\alpha = 0.001$, with $\sigma = 6$, as a result of the hydrodynamic instability. The flame profiles $\varphi(x)$ are plotted at consecutive times, from top to bottom. The spontaneous appearance of small wrinkles is clearly seen as they develop near the trough and propagate along the surface towards the crest.

flame front. This is illustrated in Fig. 8 which shows the unsteady nature of the propagation speed relative to the speed of the corresponding steadily propagating single-peak structure, and the flame profiles at four instances in time relative to the smooth uncontaminated profile. The flame profiles have been superimposed on top of each other in order to show the associated increase in surface area which explains the increase in speed. Note that the average incremental speed of the unsteady evolving flame is nearly twice the laminar flame speed.

Direct numerical simulations of the low Mach number reactive conservation equations were recently carried out by Grcar and Matalon [72] to describe the nonlinear evolution of a nominally planar flame and the consequences of the DL instability. These intensive computations are based on an adaptive mesh refinement algorithm that was extended to model detailed kinetics and variable transport [73,74]. Each calculation begins with a randomly wrinkled flame extending across the width of the region. The width of the domain of integration was treated as a parameter with the length taken five times larger. The inflow is controlled to keep the mean flame location three domain widths above the lower boundary [75]. The simulations were carried out for methane–air (equivalence ratio 0.8) with the reduced scheme DRM-19 (corresponding to 19 species and 84 reactions) derived from GRIMech-1.2 by Kazakov and Frenklach [76], along with its associated thermodynamics and transport databases. In accord with the asymptotic results, the DNS show that a flat flame results in a narrow box, and a single-peak structure spanning the entire width is obtained in wider boxes. A more detailed discussion, including flame shape and speed, appears in the referenced paper.

The experimental record reveals similar behavior. Lind and Whitson [77] performed experiments of large expanding flames propagating in lean hydrocarbon–air mixtures. The mixture was initially filled in 5–10 m radius thin plastic hemispherical bags. The plastic film tore quickly after ignition and the entire hemisphere rose vertically. The expanding flame first appeared as a blue hemisphere, but as its size increased, the surface became rough with a *pebbled* appearance. The corrugations increased in size to about 0.4–1.0 m with a finer structure superimposed. Pressure waves were barely detectable so that the flame expanded freely at essentially constant pressure. The measured propagation velocities were 1.6–1.8 times the laminar flame speed and the results were nearly the same for all five fuels studied. Bradley et al. [78] reported on measurements of large-scale explosions of methane– and propane–air mixtures carried out to radii just beyond 3 m. Self-wrinkling was observed when the flame reached a size of approximately 0.5 m, and cells as large as 40 cm were detected for both mixtures with no significant difference discerned between methane and propane.

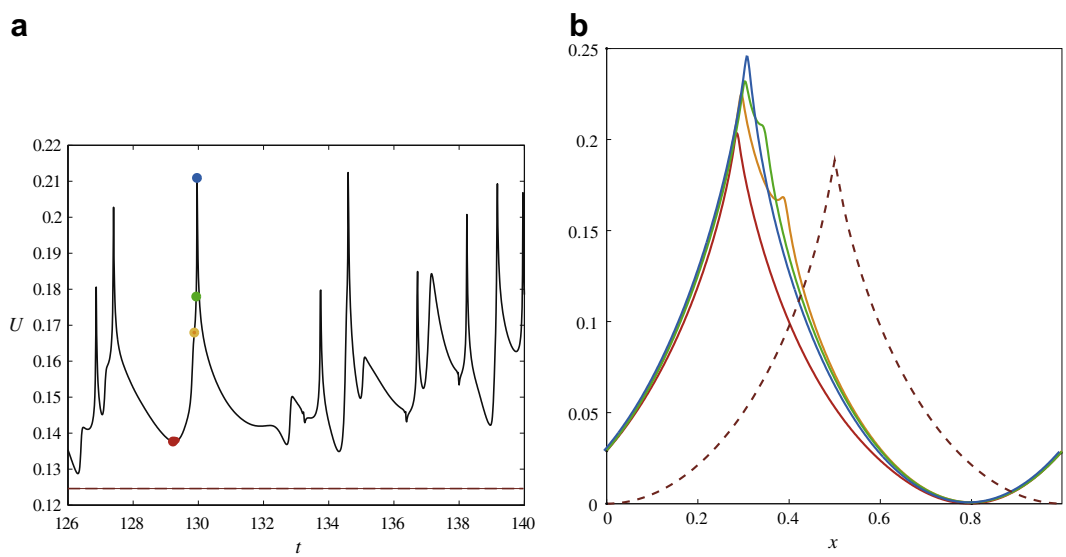

Fig. 8. The unsteady nature of the evolving flame in large domains ($\alpha = 0.0035$): (a) time-history of the (scaled) incremental increase in speed U, relative to the corresponding smooth single-peak structure (dashed line) and (b) flame profiles $\varphi(x)$ at the four marked instances, superimposed on top of each other, relative to the flame shape of a smooth single-peak structure (dashed curve). The appearance of a small wrinkle at the trough (on the right side) traveling along the flame surface is associated with the recorded increase in speed until its disappearance at the crest.

2.4. Flame propagation in channels

Another problem that involves flame–flow interaction, for which significant advances have been made, is the propagation of a curved flame in a *two-dimensional* channel. The analysis was carried out within the context of a constant-density model, namely assuming that the density is independent of temperature so that the flow field not affected by the flame may be prescribed a-priori. The transport equations were solved numerically for a one-step global reaction with finite-rate kinetics and verified against asymptotic results in special limits. The results illustrate the effects of channel width, conductive heat losses, differential diffusion and a convective flow on the combustion characteristics.

When a quiescent combustible gas mixture contained in a channel is ignited at one end, a flame propagates through the gas. Under adiabatic conditions and when the channel is sufficiently narrow, the flame is planar[2] and propagates at the laminar flame speed S_L. Results for non-adiabatic walls [79] are presented in Fig. 9, where the heat flux through the walls is assumed proportional to the temperature difference between the gas and the walls with an appropriately defined heat transfer parameter k_c. It ranges from $k_c = 0$, corresponding to adiabatic conditions, to $k_c \to \infty$, corresponding to cold isothermal walls. The graph shows the dependence of the propagation speed on heat losses for various channel widths. The parameter $\delta = d/l_f$ represents the channel width d measured in units of the laminar flame thickness. The propagation speed is seen, in general, to decrease with increasing k_c. In narrow channels, total extinction is brought about when k_c exceeds a critical value. The propagation speed at extinction is nearly 60% the value of the laminar flame speed. The flame in this limit is quasi-one-dimensional and the extinction conditions are well captured by those deduced from the asymptotic results of a planar flame with volumetric heat loss [34]. The graph illustrates that total extinction by heat loss would only occur when $\delta < 15$, implying a quenching diameter $d_q \sim 15 l_f$ which agrees with experimental observations. In wider channels burning persists even in the extreme case of isothermal walls maintained at the temperature of the fresh mixture, but the flame survives only near the center of the channel. The dead-space corresponding to the distance near the walls where the flame is quenched can be as large as $6l_f$ in accord with experimentally reported values

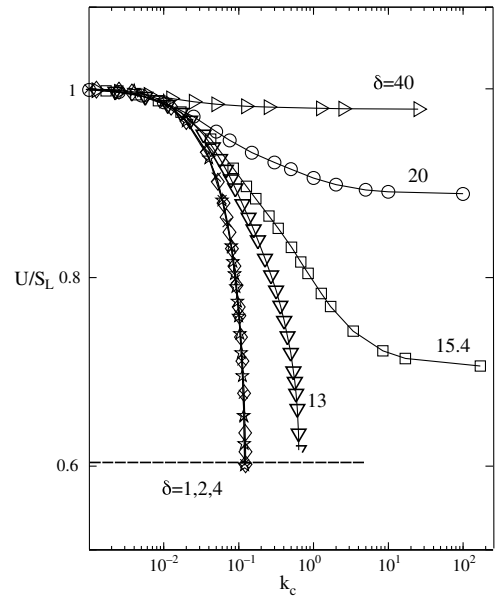

Fig. 9. Flame propagation in channels of various width $\delta = d/l_f$. The graph shows the dependence of the propagation speed as a function of conductive heat losses, represented by k_c, for different values of δ.

[80,81]. For sufficiently large values of δ, the walls have a very limited influence on the flame, which then propagates at a speed close to the laminar flame speed. Results on the quenching radius for ducts of circular cross-section and its dependence on the mixture's Lewis number were reported by Kurdyumov and Fernández-Tarrazo [82].

The effect of a convective flow on flame propagation was studied for adiabatic [83] as well as non-adiabatic [79] walls. The flow, assumed to be Poiseuille flow, can be directed from the unburned towards the burned gas or vice versa, with the flame either opposed or assisted by the flow, respectively. When opposed by the flow, the flame shape is as shown in Fig. 10(a), with its central part pointing towards the burned gas and its leading edge at the wall. At sufficiently low flow rates, the flame propagates to the left, a condition referred to as flashback. For critical flow conditions the flame is stabilized, but beyond criticality it propagates to the right, a condition referred to as blowoff. When assisted by the flow the flame is always blown off and propagates to the left. Its shape is as shown in Fig. 10(b), namely with its central part pointing toward the unburned gas and its leading edge at the centerline. The graphs in Fig. 10 illustrate the effect of the channel's width on the structure of a flame (a) opposed, or (b) supported by the flow. Each panel displays reaction rate contours in the range $0 < \omega < \omega_{max}$, with the maximum value quoted.

[2] Beyond a certain channel width the flame is curved, which is a manifestation of the hydrodynamic instability. In the present context, however, because thermal expansion is assumed negligible, this effect is absent.

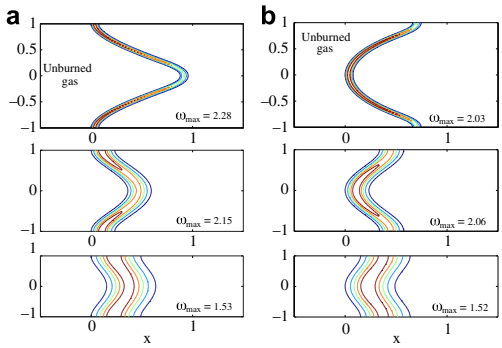

Fig. 10. Flame propagation in a channel with adiabatic walls (a) opposed by and (b) aligned with a Poiseuille flow. Shown in the figures are reaction rate contours that identify the flame shape and broadness. For the reported calculations the flame always propagate to the left and its leading edge is set at the origin. The figure illustrates the effects of channel width with $\delta = 20, 7, 3$, respectively.

The increase in flame surface area in the presence of a convective flow is associated with an increase in total burning rate. In both cases the burning rate increases but the rate of increase is slightly higher when the flame is opposed by the flow, as can be also deduced from the figure; the flame length and $\omega_{\max}$ are both larger in this case. Graphs similar to those shown in Fig. 10 were obtained for non-adiabatic walls and various flow conditions [79]. Results indicate that the flame is more vulnerable to heat losses when it is opposed by the flow than when assisted by the flow. In particular, the critical flow conditions for flame stabilization decreases with increasing k_c, implying that it is easier to stabilize a flame in the presence of heat losses as is well known from experience with burners.

Figure 10 also illustrates the effect of channel width on flame propagation under adiabatic conditions. In narrow channels, with dimensions comparable to the flame thickness, the flame is broader and flattens out, and the burning intensity decreases. The burning rate of these relatively thick flames is found to be proportional to the square of the mean flow velocity and is therefore not affected by the flow direction. And unlike thin flames, the burning rate is not affected by variations in Lewis numbers. Transverse diffusion in the preheat zone becomes comparable to axial diffusion, differential diffusion is not accentuated and the burning rate tends to the laminar burning rate for all Lewis numbers [84]. This may have an implication on turbulent combustion, suggesting that Lewis number effects may be more relevant in the flame sheet regime, where the representative turbulent length scale is larger than the flame thickness, than in the reaction sheet regime where the two length scales are comparable.

Differential diffusion has also an influence on the nature of propagation. When the effective Lewis number of the mixture is sufficiently large, which can be realized by reducing the mass diffusivity of the deficient reactant, for example by adding to the mixture an appropriate diluent, a pulsating mode of propagation is observed [85]. Near the bifurcation point the oscillations have a sinusoidal behavior and the flame propagates as a whole with an average speed that is close to the laminar flame speed. For larger values of the Lewis number, however, the flame performs relaxation oscillations characterized by intermittent long-time intervals, where the flame is nearly quenched, separated by short-time intervals where the reaction rate increases rapidly to cause a forward motion with a speed that significantly exceeds the laminar flame speed. The dependence of a convective flow and of heat loss at the walls on the onset and nature of the oscillations were also examined in [85]. Recent experimental work has revealed flame dynamics that is particular to small or "meso"-scale ducts, namely of diameter which is on the order of the flame thickness [86,87]. In these circumstances the tube wall becomes thermally active relatively fast in all of its length and conducts heat from the flame to the cold gas upstream. The competing effects of the additional heat input to the fresh mixture with the natural tendency of flame quenching at such small scales leads to flame oscillations or a periodically repeated extinction/re-ignition phenomena [88]. Attempts to theoretically describe this behavior have been presented in [87,89].

Figure 11 illustrates another phenomenon associated with differential diffusion. The sub-figures correspond to flames opposed by the flow for mixtures with Lewis numbers less than one, such as rich hydrocarbon–air (except for methane) mixtures, under adiabatic conditions. Moving from the left panel to the right the effective Lewis number decreases, or the hydrocarbon–air mixture is made richer. One observes that when the Lewis number becomes sufficiently small, the tip opens up and the opening increases when the Lewis number is further reduced. Despite the significant drop in reaction rate near the tip, there is no leakage of the deficient reactant and the flame survives along the entire interface as discussed by Kagan and Sivashinsky [90]. (Local extinction with incomplete combustion near the tip is only possible in the presence of radiative losses). Differential diffusion is responsible here for the non-uniform burning pattern along the curved flame, which is more intense along the sides than near the tip. Indeed, flame tip opening has been experimentally observed in the closely-related problem of Bunsen flames [91,26] for rich, but not lean propane air mixtures, and in numerical simulation of H_2–air Bunsen flames [92]. Finally, when the flame is supported by the flow the opening

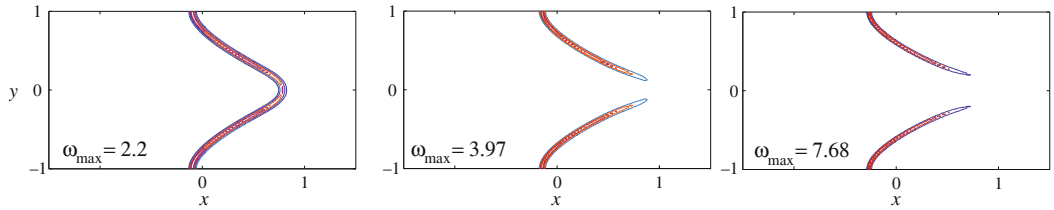

Fig. 11. Flame propagation opposed by Poiseuille flow for mixtures with a Lewis number less than one. The figure illustrates the phenomenon of tip opening when the Lewis number becomes sufficiently small.

occurs near the walls, namely at the flame's leading edge.

3. Diffusion flames

When fuel and oxidizer initially separated are brought together, mixing and reaction occur simultaneously in the reaction zone. The flame is commonly known as a diffusion flame, since diffusion is the rate-controlling process. Unlike premixed flames, diffusion flames do not have a characteristic speed; the burning rate is determined by the rate at which the fuel and oxidizer are transported from opposite sides in the right proportions to sustain the chemical reaction.

Flat premixed flames are readily established in the laboratory in a carefully controlled uniform flow kept equal to the laminar flame speed. In contrast, a steady planar diffusion flame in a truly one-dimensional setting, with one of the reactants supplied far upstream and the other far downstream, is not possible. For a steady diffusion flame to exist, one must have nonzero fluxes of fuel and oxidizer toward the flame, and this cannot be produced in a one-dimensional flow because the net flux of the reactant originating at the downstream end must remain fixed.[3] A one-dimensional model—the chambered diffusion flame—can be formulated if one of the reactants is supplied at a finite location [93,94]. In the present discussion fuel is fed from the bottom of the chamber at a constant rate, and oxidizer diffuses uniformly from the top of the chamber against the flow of products. To sustain a steady flame, conditions at the top must be maintained constant, which has been realized in the idealized-construct by a fast flowing stream of oxidizer across the exit of the chamber; see Fig. 12(a). However, it is experimentally difficult, if not impossible, to retain a uniform concentration of oxidizer at the top of the chamber and ensure the removal of product gases. This experimental difficulty has been overcome only recently by Lo Jacono et al. [95] in a novel experimental design. In this burner the oxidizer is introduced into the chamber through an array of hundreds of closely spaced hypodermic needles, spaced equally to allow the hot combustion products to escape vertically between the needles. The inhomogeneity near the exit plane of the needles is limited to a layer of thickness comparable to the needle spacing and the desired uniform conditions are realized just beyond this layer. Figure 12(b) shows a side-view photograph of a planar unstrained flame. The measured strain rates at the flame front were less than 0.5 s^{-1} and these resulted from flow non-uniformities caused by heat loss to the chamber walls [96]. These imperfections were circumvented in the latest design of the chamber [97], which has been also improved to permit better control of the magnitude and direction of the bulk flow. In the new symmetric design both reactants are supplied through hypodermic needles which allows the creation of a "pure" diffusion flame with no net flow across the flame.

The experimental realization of a planar *unstrained* diffusion flame opens the realm possibilities for the exploration of fundamental questions related to diffusion flames, and for the

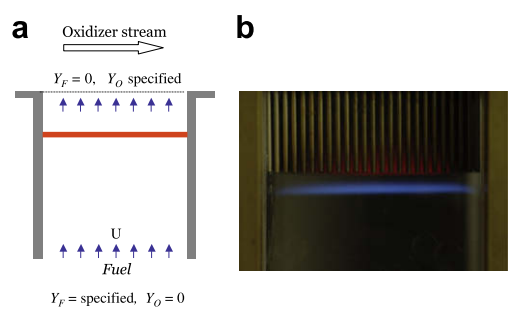

Fig. 12. The chambered diffusion flame: (a) the theoretical model, and (b) side view photograph of a planar unstrained flame sustained by a stream of CO_2-diluted hydrogen fed from the bottom with oxygen diffusing from the top against the stream. Courtesy of P. Monkewitz.

[3] Planar diffusion flames are commonly established in the flow field of two impinging jets issued from opposing nozzles. However, the underlying flow field in this configuration is two-dimensional and the flame is strained.

experimental validation of theoretical and numerical models. It provides for the first time a validation of the results on flame instabilities, which have been primarily based on the chambered diffusion flame model [98–100] as described below. In addition, the unstrained planar flame can be potentially used for experimental studies of the structure, including oxidation mechanisms and transport rates of diffusion flames in their purest form, *i.e.* where mass transport is by diffusion only, and where the flames are free of the impeding effects of fluid dynamic strain, or curvature.

3.1. Reaction sheet model

The mathematical description of diffusion flames dates back to Burke and Schumann [2], who introduced the infinitely fast-chemistry limit. They hypothesized that chemical reaction is confined to a sheet that separates a region where there is fuel but no oxidizer, from a region where there is oxidizer but no fuel. Fuel and oxidizer flow into the reaction sheet in stoichiometric proportions, a condition that along with the requirement of complete consumption of reactants determines the combustion field including the shape and location of the reaction sheet, or stoichiometric surface. The latter depends on the fuel and oxidizer concentrations at the supply boundaries, or initial mixture strength ϕ, and on the diffusivity of the reactants, or Lewis numbers Le_F and Le_O. Formally, the Burke and Schumann solution corresponds to the limit $\mathbb{D} \to \infty$, where $\mathbb{D}$ is the Damköhler number [101–103]. The thin reaction zone is "passive" and except for smoothly connecting the discontinuity in slopes it imposes no restrictions on the solution.

A significant advancement in the analysis of diffusion flames is due to Liñán [104] who reexamined the structure of the reaction zone in the context of a counterflow flame assuming that both the Damköhler number and the activation energy parameter are large but related in an appropriate distinguished limit. The structure of the reaction zone now depends on the reduced Damköhler number δ, that measures the intensity of the chemical reaction, or the departure from equilibrium conditions. Integration across the reaction zone shows that the reaction–diffusion balance is sustained only when $\delta \geqslant \delta_c$. The critical value δ_c determines the lowest possible value of the Damköhler number, $\mathbb{D}_{ext}$, that corresponds to flame extinction.[4] For the counterflow flame $\mathbb{D}_{ext}$ determines the maximum possible strain rate above which extinction occurs; for a spherical flame it determines the critical size below which extinction occurs. Figure 13 shows the relevant part of the S-response curve that characterizes the solution which extends from complete combustion down to extinction, or $\mathbb{D}_{ext} \leqslant \mathbb{D} < \infty$. The graph illustrates the dependence of the flame temperature T_f on $\mathbb{D}$. As $\mathbb{D} \to \infty$ the curve asymptotes to the Burke–Schumann solution corresponding to complete combustion with the flame temperature equal the stoichiometric temperature T_s. The intensity of the reaction diminishes when $\mathbb{D}$ is reduced; as a consequence the flame temperature drops and there is significant leakage of unburned reactants through the reaction zone. Extinction occurs when the leakage becomes excessive and the flame temperature drops significantly below T_s.

In a recent development that parallels Liñán's seminal work, Cheatham and Matalon [98] derived a general asymptotic formulation that is not restricted to any particular geometry and allows for non-unity Lewis numbers. Chemical reaction is confined to a sheet which is near, but does not necessarily coincide with the stoichiometric surface. The internal structure of the reaction zone depends on the auxiliary parameters δ and γ. The parameter δ is associated with the intensity of the reaction and controls the total amount of heat generated in the reaction zone. The parameter γ determines the amount of heat needed on each side of the sheet to sustain combustion. At any point along the sheet these parameters are determined by the local flow, temperature and concentration conditions. Numerical integration of the reaction zone provides through asymptotic matching jump conditions in terms of γ and δ, that permit the determination of the combustion field as well as the shape and location of the sheet itself. These conditions include the extent of fuel and oxidizer leakages Y_F^* and Y_O^*, which for $\gamma > 0$ are proportional to the functions S_1 and S_2 shown in Fig. 14; for $\gamma < 0$ the roles of S_1 and S_2 are interchanged. Both functions vanish as $\delta \to \infty$, a limit corresponding to the Burke–Schumann solution for which $Y_F^* = Y_O^* = 0$. The amount of fuel and/or oxidizer leaking through to opposite sides of the reaction zone increases as $\mathbb{D}$, which is related to the parameter δ decreases, with the fuel preferred[5] when $\gamma > 0$ and oxidizer when $\gamma < 0$. The leakage of one of the reactants becomes excessive as $\mathbb{D} \to \mathbb{D}_{ext}$, and a further reduction in $\mathbb{D}$ leads to extinction. Explicit formulae for S_1

[4] It is often convenient to use the mixture fraction $Z(\mathbf{x}, t)$ as one of the coordinates [35]; the extinction condition then depends on the scalar dissipation rate $|\nabla Z|^2$.

[5] The preferred reactant that leaks through the reaction zone in this one-step activation energy asymptotic approximation does not always agree with experiments. For hydrocarbon–air flames rate–ratio asymptotic approximation of reduced chemistry models provides better results, but such analysis have been limited at the present to one-dimensional flame structure studies.

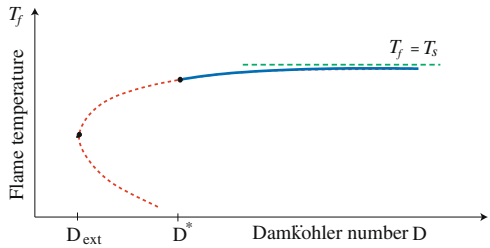

Fig. 13. The dependence of the flame temperature of a diffusion flame on the Damköhler number (upper portion of the familiar S response curve). The turning point $\mathbb{D} = \mathbb{D}_{ext}$ corresponds to flame extinction. The dashed segment corresponds to states that, depending on the Lewis numbers and the initial mixture strength, may be unstable; the critical value D^* is the marginally stable state.

and S_2, which represent the numerical results with sufficient accuracy, are presented in the referenced article, and were used in a number of studies to facilitate the analysis circumventing the need to integrate the structure equation numerically for every set of conditions [98–100,105].

The entire formulation is a nonlinear free-boundary problem for the combustion field, including the location of the reaction sheet, with conditions that permit incomplete combustion with different degrees of reactant consumption. It is therefore appropriate for analyses of multidimensional and time-dependent problems. The theory predicts local extinction along the sheet and hence the possible creation of holes. However, it does not provide information about the evolution of holes where combustion is presumably governed locally by the characteristics of edge flames, a topic that will be further discussed below. Modeling efforts on the dynamics of holes have been recently initiated for the simple case of circular flame holes [106], but more work is needed in this area.

3.2. Flame instabilities

Studies of flame instability have been predominantly concerned with premixed combustion, but there is ample experimental evidence of instabilities in diffusion flames[6] The first known study is due to Gardside and Jackson [107] who observed that the surface of a hydrogen–air jet flame often comprises of triangular cells in the shape of a polyhedron. Later Dongworth and Melvin [108] observed that under certain conditions the straight base of a hydrogen–oxygen diffusion flame on top of a splitter-plate burner becomes cellular. Similar observations were reported in recent studies of a Wolfhard–Parker burner [109], jet [110,111] and counterflow [112] diffusion flames. A common feature of all these observations is that the instability results at sufficiently high flow rates, or at near-extinction conditions. Another feature is that cells form when at least one of the Lewis numbers is sufficiently small; for example, in hydrogen–oxygen flames diluted in CO_2, N_2 or Ar, in which case $Le_F \sim 0.33$–0.35, but not when diluted in He because Le_F is near one and Le_O is quite large. Similarly they form in methane–air diluted with SF_6, in which case $Le_F, Le_O \sim 0.4$–0.5, but not when diluted in He or N_2 because the Lewis numbers are again near or larger than one. The propensity of diffusion flames to form cells is increased when decreasing the initial mixture strength ϕ, defined as the fuel-to-oxidizer mass supplied in their respective streams normalized by their stoichiometric proportions [111].

Another form of instability that appears in non-premixed systems is spontaneous flame oscillations. Oscillations were observed in condensed-phase fuels [113], candle and large suspended fuel droplets in a microgravity environment [114,115], jet diffusion flames [116], spray flames [117] and flame spreading over liquid beds [118,119]. The nature of the oscillation in each of these experiments is quite different: The droplet flame exhibits radial oscillations and the jet flame expands and contracts as a whole during a cycle. For the microgravity candle flame the edge is seen to move back and forth along the hemispherical flame surface. In flame spreading the oscillations are also primarily seen near the edge, decaying along the trailing diffusion flame.

The underlying hydrodynamics in the reported experiments is generally non-trivial, involving multidimensional shear and strained flows that produce nonuniform conditions over the flame surface and complicate comparison with theoretical predictions. The recent realization of an unstrained planar diffusion flame provides the opportunity to investigate flame instabilities in their simplest form. Results on the onset of cellular flames in a CO_2-diluted hydrogen–air system have been already reported [95]. The oxidizer concentration is set by controlling the oxygen injected from the hypodermic needles, and the hydrogen mass fraction in the fuel stream is gradually decreased by increasing the CO_2 concentration in the incoming stream. The flame appears first flat, but when reducing the fuel concentration below a critical value it takes on a cellular appearance as shown in Fig. 15. The cellular structure is retained when the hydrogen concentration is continuously lowered down to a limiting value where flame extinc-

[6] The focus here is on intrinsic instabilities associated with the combustion process itself. Combustion studies that favor an unstable underlying flow that promotes mixing are often of interest, but are not considered here.

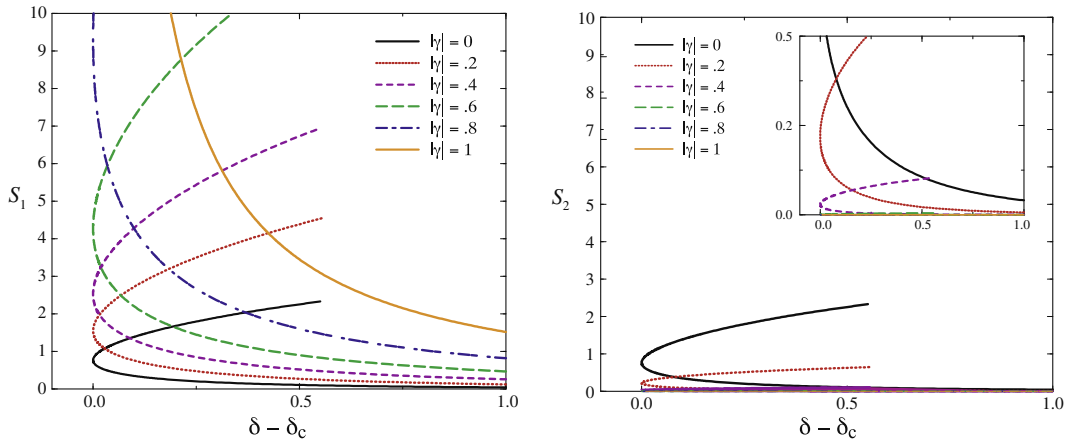

Fig. 14. The functions S_1 and S_2 associated with the reactant leakage through the reaction sheet as a function of the parameters δ and γ. The degree of reactant consumption depends on δ with complete consumption occurring when $\delta \to \infty$. For $\gamma = 0$ there are equal amounts of fuel and oxidizer leaking though the sheet from opposite directions; more fuel leaks through the sheet when $\gamma > 0$ and more oxidizer leaks through the sheet when $\gamma < 0$.

tion occurs. Cellular flames are observed over a significant range of H_2 concentration, corresponding to initial mixture strengths $\phi < 1$ and for Lewis numbers in the range $Le_O \sim 0.97$–1.33 and $Le_F \sim 0.22$–0.29. This range narrows when increasing ϕ, and for $\phi > 1$ the systematic reduction of hydrogen concentration led directly to extinction. The observed cells at the transition are ~ 0.7–1 cm wide, with the smaller size seen when the transition occurred near the extinction limit. More recent results by Robert and Monkewitz [97] are reported in a paper presented in this symposium.

3.3. Cellular and oscillating flames

Stability analyses of diffusion flames have predominantly used a planar flame as a basic (unperturbed) state [98–100,120–123]. Although the models used by various investigators differ slightly by the boundary conditions imposed, these minor differences do not affect the results in any significant way. The most elaborate results are obtained using the reaction sheet formulation, further assuming that the density is constant in order to filter out hydrodynamic disturbances. An explicit dispersion relation can be derived but due to its transcendental nature, finding roots for the growth rate in the complex plane requires substantial numerical computations [98,99]. The free-boundary eigenvalue problem was also solved numerically [100] and together with the analytical results yielded comprehensive results that included the critical conditions for the onset of instability and a characterization of the possible patterns that are likely to be observed beyond the instability threshold.

The Burke–Schumann solution of complete combustion is unconditionally stable. If the planar flame is slightly perturbed, the requirement of complete reactant consumption forces the reaction sheet back to its planar form which, for the given parameters, is the only stoichiometric surface. Instabilities can only exist when there is appreciable reactant leakage, or when the Damköhler number is sufficiently low (the dashed segment in Fig. 13), *i.e.* in the range $\mathbb{D}_{ext} \leqslant \mathbb{D} < \mathbb{D}^*$. Since the characteristic flow time for the chambered flame is l_D/U, where $l_D = \mathscr{D}_{th}/U$ is the diffusion length and U is the bulk flow velocity, this corresponds to relatively high flow rates or near-extinction conditions. The marginally stable state $\mathbb{D}^*$ and the nature of the instability when $\mathbb{D}$ is

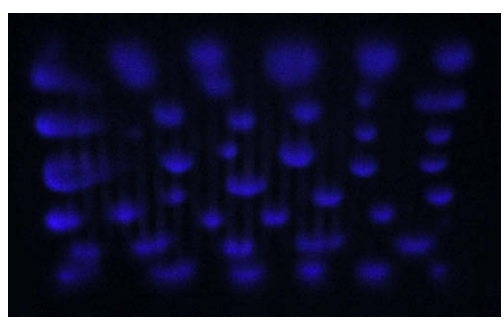

Fig. 15. Photograph (taken in an oblique angle) of a cellular diffusion flame. The nominally planar flame was sustained by a stream of CO_2-diluted hydrogen fed from the bottom with oxygen diffusing from the top against the stream. The cellular flame is obtained by reducing the H_2 concentration in the feeding stream. Courtesy of P. Monkewitz and E. Robert.

reduced below $\mathbb{D}^*$ depend on the mixture composition and on the mobility of the reactants.

The diagram in Fig. 16 shows the various patterns that are likely to be observed at the instability threshold in the Lewis numbers parameter plane. The dashed vertical line separates regions of relatively lean (to the left) and rich (to the right) mixtures, and shifts to the right as the initial mixture strength ϕ decreases. Stationary cells occur in the region below the solid curve shown in the stability diagram that encompasses the square $0 < Le_F < 1,\ 0 < Le_O < 1$. This region stretches out to include a wider range of Le_O when reducing ϕ, implying that leaner mixtures are more susceptible to cellularity. Characteristic cell size at the onset of the instability are in the range $\lambda \sim 3$–$12 l_D$, depending on the mixture strength and the flow rate [100]. For conditions associated with the CO_2-diluted hydrogen-oxygen chambered flame, the predicted cells are 0.5–2 cm wide. The smaller cells are expected near the extinction limit, or when $\mathbb{D}^*$ is sufficiently near $\mathbb{D}_{ext}$, in which case the analysis must be reconsidered to incorporate small wavelength perturbations comparable to the reaction zone thickness that evolve on the fast chemical reaction time [121]. The theoretical predictions of the critical conditions, including flow rate, mixture strength and Lewis numbers, for the onset of cellular flames as well as the characteristic cell size are in agreement with the experimental record.

Planar pulsation occurs in the upper right corner of the stability diagram and is associated with sufficiently large Lewis numbers. Since the dashed vertical line shifts to the left as ϕ increases, oscillations are favored in relatively rich mixtures. Planar pulsation for the chambered diffusion flame

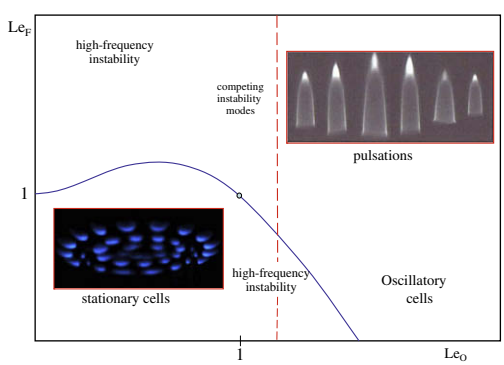

Fig. 16. Typical stability diagram illustrating the various patterns that are likely to be observed at the onset of an instability of a nominally planar diffusion flame. The dashed vertical line separates regions of relatively lean (to the left) and rich (to the right) mixtures, and shifts to the right as ϕ decreases; the solid curve which always includes the point $Le_F = Le_O = 1$ stretches out as ϕ decreases.

has been recently observed in rich mixtures with one of the Lewis numbers sufficiently large. The conditions concerning the onset of oscillations appear also commensurate with other experimental observations [116]. Oscillations were observed for propane-air jet flames diluted in nitrogen when $\phi \geqslant 1.32$, in which case Le_F varies from 1.1 to 1.8 with $Le_O \approx 1$, but were not observed in the relatively lean system with $\phi \leqslant 0.76$ despite the large-fuel Lewis number that resulted from the N_2 dilution. These conditions fall within the predicted regions [100] for the corresponding value of ϕ. To examine the development beyond the instability threshold, Cheatham [124] carried out a bifurcation analysis and obtained amplitude equations that lead to either time-periodic solutions corresponding to sustained oscillations, or unbounded solutions that imply premature extinction or blowoff. The dependence of these results on the parameters was discussed in [125].

Although cellular structures and flame oscillation are the predominant forms of instability, other possible patterns, such as oscillating cells or competing modes of comparable and/or disparate scales, may exist in the transition regions between various domains or for extreme values of the parameters. Mixed modes of instability were reported, for example, in jet diffusion flames in the form of travelling, or rotating cells [110] and long cells were reported in a recent study of non-premixed tubular flame [126].

The mathematical consequence of non-unity Lewis numbers on the combustion field is that it yields temperature and concentration profiles that are non-similar. Physically non-unity Lewis numbers represents the competing effects of heat conduction away, and diffusion of fuel and oxidizer toward the reaction zone. Volumetric heat loss, which affects the temperature but not the concentration fields, also generates non-similar profiles that can promote diffusive-thermal instabilities. In the presence of gas-phase radiation, for example, it was found that flame oscillation can be triggered by appreciable heat losses even when the Lewis numbers are identically equal to one [127], and that the instability is enhanced by heat losses when the conditions already favor oscillations [99]. Indeed, oscillations were observed in jet flames when a fraction of the fuel supplied was introduced in the form of liquid droplets that extract heat from the gaseous phase for their vaporization [117]. Under similar conditions, but with the droplets content replaced by fuel vapor, the flame was stable. Further compelling evidence of the role of heat loss was provided when water droplets were injected instead of the fuel droplets leading to oscillations of similar frequencies.

The effect of thermal expansion on the stability of a planar diffusion flame has been considered only recently by Metzener and Matalon [128]. Because thermal expansion is not the driving

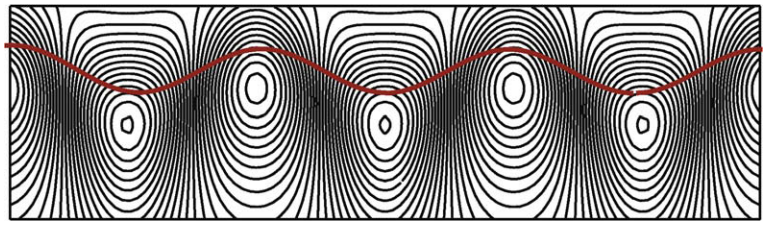

Fig. 17. The vorticity field in the vicinity of a cellular flame at the instability threshold, calculated for a density ratio ≈ 3. The solid curve denotes the reaction sheet. The concentrated vortical motion near the crests and troughs enhances the transport of fuel to these regions, thus sustaining the cellular structure.

mechanism for instability, the general characterization described above remains qualitatively correct even when realistically accounting for density variations. In particular, the Burke–Schumann state of complete combustion remains unconditionally stable. The main effect of thermal expansion is varying the degree of instability and consequently shifting the marginal state $\mathbb{D}^*$. In particular, it is found to play different roles on the various mode of instability. The growth rate in the parameter range corresponding to cellular flames increases with increasing thermal expansion, implying that thermal expansion acts to further destabilize the flame. The induced flow resulting near a cellular structure involves regions of concentrated vorticity upstream of the reaction surface with extrema near the crests and troughs (Fig. 17). The vortical motion enhances the transport of fuel to these regions, which sustains the corrugated structure. The onset of cellular flames occurs at a value $\mathbb{D}^*$ significantly larger than that predicted by the constant-density model, which implies that a wider range of physical states are susceptible to cellularity. On the other hand, the growth rate associated with planar pulsation decreases when increasing the thermal expansion parameter. Since the flame remains planar and its back and forth motion is towards denser fluid, thermal expansion has a damping influence on the oscillations. The marginal state $\mathbb{D}^*$ is significantly lower than the value predicted by the constant-density model, which implies that a narrower range of physical states are susceptible to pulsation. Overall, although thermal expansion has a marked influence of the dynamics of diffusion flames, it does not play a crucial role as it does for premixed flames.

4. Edge flames

The planar deflagration and chambered diffusion flames are two fundamental structures that have been traditionally used to gain understanding into the more complex flame behaviors that occur in real applications. They constitute two extremes in which fuel and oxidizer are either well mixed before introduced into the combustion chamber, or supplied from different origins with mixing and combustion taking place simultaneously. The edge flame is another fundamental structure, albeit a two-dimensional one, which is associated with partially premixed conditions and thus has characteristics of both premixed and nonpremixed flames. A representative edge flame formed in the wake of splitter plate separating two parallel streams, one containing fuel and the other oxidizer, is shown in Fig. 18(b). Partial mixing occurs behind the tip of the plate and combustion takes place in the stratified medium, once the mixture is ignited. The edge flame has a tri-brachial structure (also referred to as a triple flame) which consists of a highly curved premixed flame with lean and rich branches, and a diffusion flame trailing behind consuming the remaining unburned reactants.

Edge flames occur in variety of situations. At low speeds, a fuel jet injected into an oxidizer environment burns, when ignited, along a diffusion flame attached to the nozzle. At high flow rates, the flame is stabilized further downstream within the jet. Lifting the flame base off the nozzle has the advantage of avoiding thermal contact

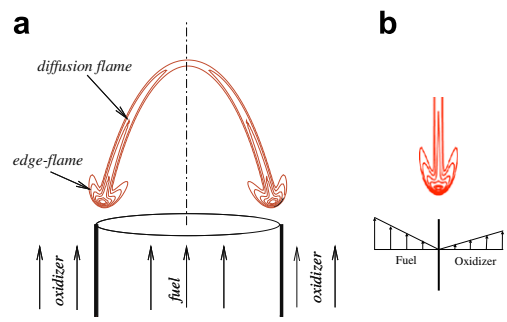

Fig. 18. (a) Lifted jet diffusion flame, calculated for stoichiometric conditions and $Pe \equiv RU/\mathscr{D}_{\rm th} = 5$, where Pe is the Peclet number based on the radius R of the nozzle and a representative gas velocity U. (b) A sketch of the local structure of the edge flame near the nozzle illustrating its tri-brachial structure.

between the flame and the rim and enhancing mixing. The lifted flame has an edge as shown in Fig. 18, which has been also observed experimentally [129,130]; see also the more recent review article by Chung [131]. Edge flames also occur near holes poked in turbulent diffusion flames as a result of excessive thermal gradients or high strain. Through the hole, fuel and oxidizer come in contact and a stratified pocket of premixed gas varying from lean to rich conditions is formed. When thermal gradients increase well above the quenching limit through turbulent fluctuations, the gas mixture can ignite and propagate in the form of an edge flame. The edge flame can either propagate inwards into the pocket of fresh gases, reestablishing the flame and reducing the size of the hole, propagate outwards, spreading extinction to other parts of the flame, or "holding" in some way those parts of the flame that did not extinguish. This has been observed experimentally near the axisymmetric hole created in a counterflow diffusion flame [132–134]. An edge flame has been also observed in the weakly nonuniform straining field created in an opposed slot-jet burner with nonparallel jet exits [135]. The diffusion flame is sustained in the region where the strain rate is sufficiently low but is quenched, and thus terminates with an edge, when the strain rate is above a critical value.

4.1. Mathematical description

The earliest analytical attempt to describe the structure of an edge-flame was made by Ohki and Tsuge [136], but the interest in such problems started a few years later and have focused on flame propagation in weakly stratified mixtures [137–140]. Analytical considerations were based on perturbation of the planar premixed flame, neglecting the effect of thermal expansion. From these studies expressions for the speed of the edge-flame were obtained and the dependence on the imposed concentration gradient was identified. A one-dimensional model that describes the transition between burning and non-burning states along the axis of a flame sheet has been proposed as a paradigm of an edge flame [141], providing results that illustrate some characteristics of observed edges. However, an edge flame is an intrinsically two-dimensional configuration and the one-dimensional setting requires modifying the reaction term in the governing equations in order to emulate transverse diffusion that is an essential element in the physical description. Recent advances have, therefore, relied on numerical computations.

4.2. Edge flames in a mixing layer

Numerical study of a model problem that involves an edge flame in a non-premixed system, has been carried out by Buckmaster and Zhang [142]. A flat diffusion flame is assumed to exist along an axis, with fuel and oxidizer supplied at two opposing ends in the transverse direction. The edge results from cutting-off the fuel supply arbitrarily at a finite position. The calculations illustrate the onset of spontaneous oscillations in mixtures for which the fuel Lewis number is sufficiently larger than one (the oxidizer Lewis number was assumed equal to one) when the Damköhler number is sufficiently low; see also [143] and the references therein.

The more realistic problem of an edge flame in a mixing layer was treated by Kurdyumov and Matalon [144–147]. Combustion occurs downstream of the trailing edge of a plate, held at a constant temperature, and separating fuel and oxidizer streams. The earlier work assumes that the streams, of equal and constant velocities, remain unperturbed by the flow. In the more recent results the flow field is calculated based on the Navier–Stokes equations[7] with upstream profiles that emulate the local behavior of the boundary layers near the trailing edge of the plate, as illustrated in Fig. 18(b). The velocity gradients on either side of the plate are assumed equal and characterized by U/l_{BL}, where U is a representative velocity and l_{BL} is the boundary layer thickness. The flame standoff distance depends primarily on the flow rate, or on the Damköhler number $\mathbb{D}$ which is inversely proportional to U^2. At low flow rates (large $\mathbb{D}$) the edge is attached to the plate. It lifts off when the flow rate increases and stands at a well-defined distance from the tip of the plate. The position $(x_{\mathrm{w}}, y_{\mathrm{w}})$ depends on the mixture composition, or the initial mixture strength, and the mobility of the reactants, or the Lewis numbers. When there is no preferential diffusion, $Le_{\mathrm{F}} = Le_{\mathrm{O}}$, and when the reactants are in stoichiometric proportions, $\phi = 1$, the edge flame is symmetric with respect to the axis of the plate. Otherwise it leans towards the fuel or oxidizer sides; for $\phi > 1$ with $Le_{\mathrm{F}} = Le_{\mathrm{O}}$, for example, the edge flame is on the oxidizer side with the trailing diffusion flame nearly aligned with the stoichiometric surface.

Representative solutions calculated for equal and unity Lewis numbers, but with $\phi = 5$, are shown in Fig. 19. A comparison of the two figures illustrates the effect of thermal expansion, which was neglected in (a) but fully accounted for in (b). In the first case the streamlines remain parallel

[7] Although the boundary layer approximation can be used to describe the mixing process between the two streams, there is a region near the tip of the plate where the flow field must satisfy the full Navier–Stokes equations. This region is embedded in the lower deck of the much larger triple deck region that describes the flow near the plate's trailing edge.

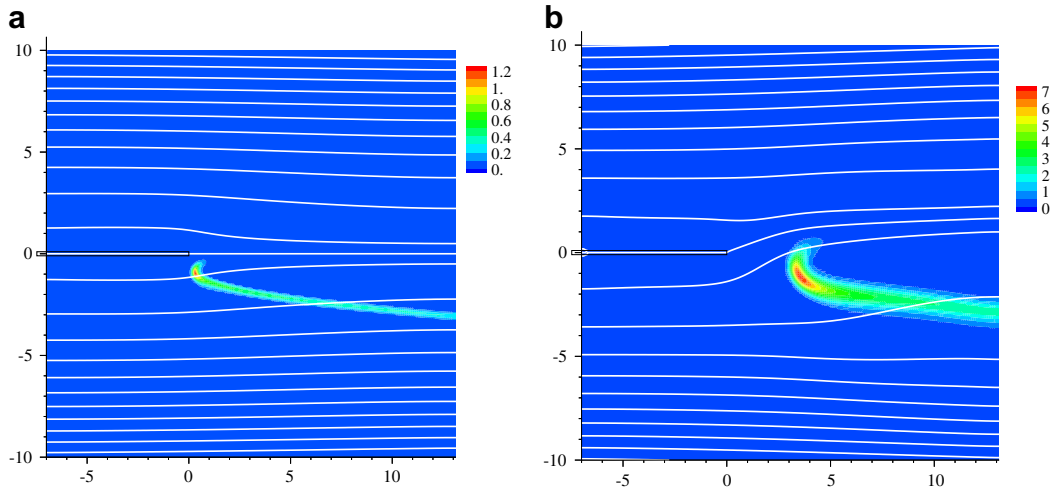

Fig. 19. An edge flame stabilized in a mixing layer calculated for a fixed value of $\mathbb{D}$, $Le_F = Le_O = 1$ and $\phi = 5$, with thermal expansion neglected in (a) but fully accounted for, in (b). In (b) the thermal expansion parameter was taken as $\sigma = 5$. The graphs display reaction rate contours that identify the flame shape and standoff distance and streamlines that show their displacement when thermal expansion has not been neglected.

and symmetric with respect to the axis. The change in slope at the station $x = 0$ is associated with the acceleration experienced by the fluid elements entrained into the mixing layer. The flame leans towards the oxidizer side, as expected, with its edge located very close to the tip of the plate and the trailing diffusion flame aligned (approximately) with the stoichiometric surface. Thermal expansion causes a deflection of the streamlines crossing the premixed segments and a lateral displacement of the streamlines near the diffusion flame. At sufficiently far distances the flow is no longer affected by the flame and becomes parallel again. The edge is seen located further downstream when compared to the case displayed in (a), but this is simply a scaling issue as discussed in greater detail in [147].

The most comprehensive results on flame dynamics were obtained using a constant-density model [146], which consists of solving the transport equations with the prescribed flow shown in Fig. 19(a). The steady-form of the equations were integrated first in order to obtain all possible solutions—stable and unstable. The time-dependent version was then solved to determine whether a given state is stable or not. Figure 20 shows results corresponding to a wide range of Lewis numbers assumed equal for the fuel and oxidizer, and denoted by Le, with $\phi = 1$. For these conditions the flame remains symmetric with respect to the axis and its position is characterized solely by x_w. The graph displays the standoff distance of the edge flame as a function of $\mathbb{D}$. The curves are generally multi-valued; for large values of Le the turning point occurs at a large x_w that is outside the domain of integration. The turning point

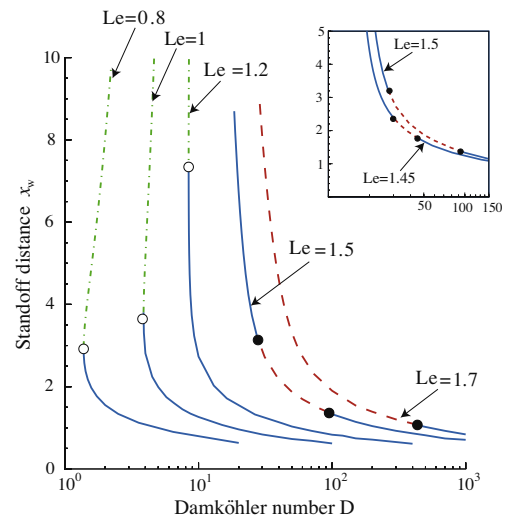

Fig. 20. The dependence of the standoff distance of the edge flame on the Damköhler number for distinct values of the Lewis numbers assumed equal (denoted by Le). The dashed–dotted segments correspond to unstable states; the symbol ∘ marks the turning point, or blowoff conditions. The dashed segments correspond to oscillatory states and the symbol • marks the marginally-stable states.

marked by the symbol ∘ identifies the lowest possible Damköhler number $\mathbb{D}_{bl}$ for flame stabilization. When reducing $\mathbb{D}$ below $\mathbb{D}_{bl}$ or, equivalently, when increasing U the flame is blown off by the flow. The upper branch (dashed–dotted segments) correspond to unstable states that cannot be realized physically. We note

that $\mathbb{D}_{bl}$ is significantly lower for mixtures with small values of *Le*. It is easier, therefore, to stabilize a hydrogen flame burning in air near the tip of the plate than it is to stabilize a hydrocarbon flame. The hydrogen flame remains near the plate when increasing the flow rate to sufficiently high values, conditions under which the hydrocarbon flame would not have survived; the flame would have been already blown off with a much lower value of the flow rate. We also note that a hydrocarbon flame can be lifted to sufficiently large distances in contrast to a hydrogen flame that can only be stabilized in the vicinity of the plate [129].

The dashed segments of the response curves in Fig. 20, seen primarily for relatively large values of *Le*, correspond to unstable states. The instability here is associated with flame oscillation. At large $\mathbb{D}$ the flame is close to the plate and is stabilized by heat loss to the plate. When $\mathbb{D}$ is lowered below a critical value, $\mathbb{D}^*$, the flame undergoes spontaneous oscillations with the edge moving back and forth along the axis dragging with it the trailing diffusion flame. It is primarily the edge of the flame that oscillates; the oscillations are weakened further downstream along the diffusion flame sheet and are damped at sufficiently large distances. For $Le = 1.7$, the onset occurs when $\mathbb{D}$ is reduced just below $\mathbb{D}^* \approx 450$ and the oscillations persist for all values of $\mathbb{D} < \mathbb{D}^*$ (within the computation domain). For the lower value $Le = 1.5$, the oscillatory states are limited to the range $\mathbb{D}_* < \mathbb{D} < \mathbb{D}^*$; the edge flame is re-stabilized when reducing $\mathbb{D}$ below $\mathbb{D}_*$ or, equivalently, by increasing the flow rate to a sufficiently high value. Stabilization here is apparently achieved at a point x_w where the edge speed is balanced by the gas velocity just in front. The time evolution of slightly perturbed states, corresponding to values of $\mathbb{D}$ just ahead, behind and within the unstable range, is shown in Fig. 21. After an initial transient the solution evolves either to the steady state or to a limit cycle. The re-stabilization phenomenon has been examined in more details in a similar problem—the flat porous-plug burner flame—which, because of its simplicity, is amenable to a complete stability analysis [148]. One also sees in Fig. 20 that the range of oscillatory states shrinks as *Le* is reduced below 1.5 and disappears when it is less than 1.45. Oscillations are not observed in mixtures with *Le* less than a critical value near one. Finally we note that when the combustion field is non-symmetric, the oscillations are associated with both coordinates varying periodically in time, and the edge-flame moves back and forth along a surface that coincides approximately with the stoichiometric surface [145].

Volumetric heat loss, such as radiative losses, have a significant effect on the combustion field. The edge flame stands farther away from the tip when heat losses are accounted for, than it would under adiabatic conditions, and the trailing diffusion flame is of finite extent. More importantly, heat losses can trigger oscillations under conditions when, in their absence, the flame is stable. For example, when the losses are appreciable the flame may undergo spontaneous oscillations [144] even in the absence of differential diffusion (*i.e.* when $Le = 1$).

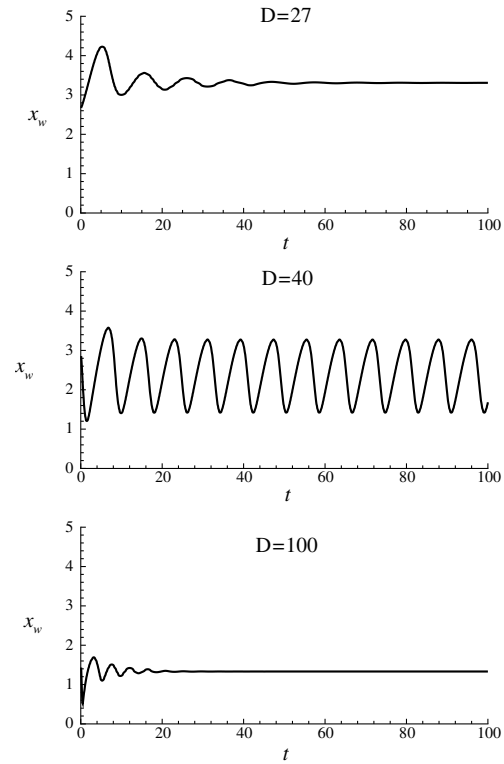

Fig. 21. Time evolution of the edge flame position for three values of $\mathbb{D}$, showing the evolution to a stable state when $\mathbb{D} = 100$, onset of oscillations when $\mathbb{D} = 40$, and re-stabilization when $\mathbb{D} = 27$. Calculated for $Le = 1.5$.

As a final comment we note that edge flames also occur in premixed systems. The edge forms in front of a relatively sharp temperature gradient, with the trailing wings supported by a straining field [149–153]. This has been observed experimentally in the weakly nonuniform straining field created in an opposed slot–jet burner with nonparallel jet exits [154]. The flame is sustained in the region where the strain rate is sufficiently low but is quenched when the strain rate is above a critical value; the edge-flame describes the transitional state between the burning and non-burning states.

5. Concluding remarks

Because of the high cost of Direct Numerical Simulations, limitations in memory and speed of available computers, theoretical and numerical

modeling will remain at least in the foreseeable future essential tools for extending our understanding of combustion and addressing the challenging problems of increasing efficiency and reducing pollutant emissions. The techniques of asymptotic analysis, which systematically exploit the various length and time scales involved in a process, are the primary tools available to derive systematic and reliable models of flames. The hydrodynamic theory of premixed flames, for example, was shown capable of describing the propagation of multi-dimensional flames with results that compare favorably with experimental observations. Similarly, the reaction sheet model was shown capable of describing time-dependent and multi-dimensional diffusion flames and providing stability results over a wide range of parameters, including thermal expansion. In either case, the theory can be extended to accommodate for more complex reaction schemes and for holes created on the flame or reaction surface, which at the present seem to be its main drawbacks.

The paper also illustrates that limiting behaviors, when carried out systematically, can provide valuable predictions that should not be discarded a-priori based only on physical intuition. The results of the MS equation, valid for weak thermal expansion, remain valid even when density variations are accounted for. These were consequently used in guiding and validating the more intense Direct Numerical Simulations. The wealth of results obtained under the constant-density assumption for flame propagation in channels and for edge flames share many characteristics of real flames and is a source of physical understanding of the complex flame-flow interactions that occur in combustion systems. Attempts to reexamine these problems while allowing for density variations are surfacing only now, with results that appear consistent with the constant-density predictions at least in some range of the parameters.

Theoretical results have been often used to guide experimental efforts. The linear relationship between flame speed and stretch for weakly stretched flames, has been tested in many circumstances and used in correlating experimental data in order to determine the Markstein number that quantifies the effects of strain and curvature. A more recent example is the experimental realization of a planar unstrained diffusion flame which has been motivated by stability predictions that have been based on the idealized chambered diffusion flame model.

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